

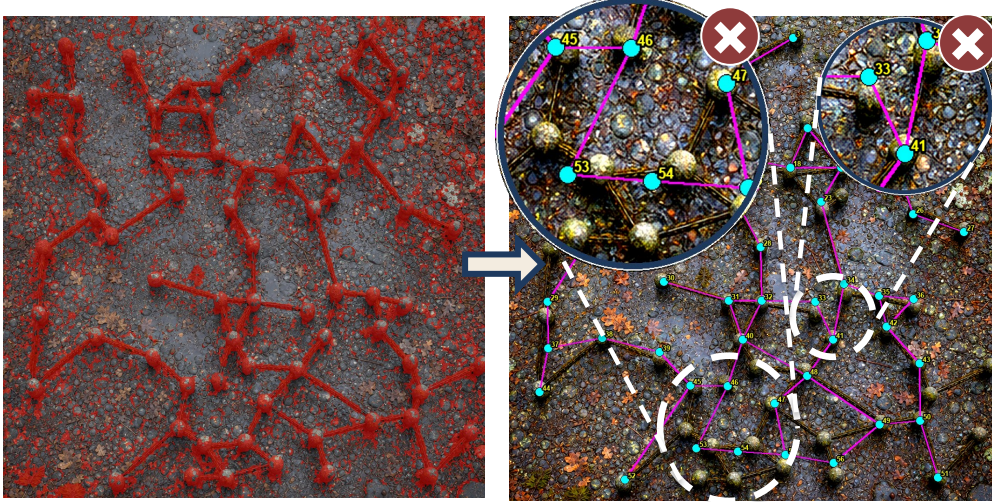
Failure 1. Tools are not robust on realistic textures.

Bounded Face Counting



How many distinct enclosed regions are visible?

Agent's Analysis Process



(a) HSV-ish mask (b) Agent's annotated image

Reasoning & Answer

GPT5.5's reasoning excerpt:

"...cobblestone cracks are noisy but separable..." presented with Figure (a)

"The Euler check gives 56 bead vertices, 67 strand edges... so $67 - 56 + 1 = 12$." presented with Figure (b)

Analysis:

The HSV-ish **mask is noisy on the realistic texture**. From it, nodes and edges are mis-recognized (56/68 nodes, 67/77 edges), so Euler over-counts (12 vs 10).

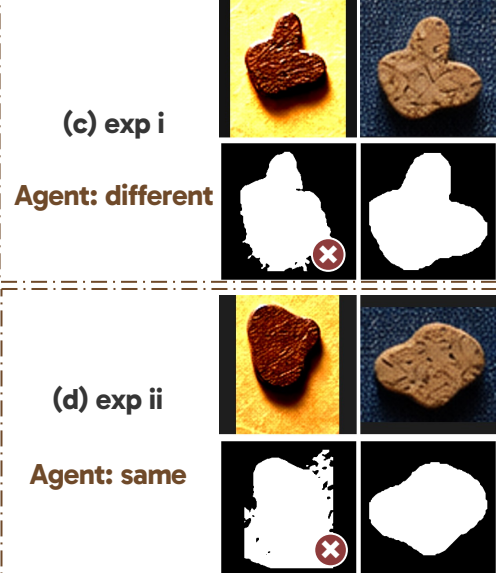
Truth: 10
Agent: 12

Contour Difference Spotting



How many silhouette pairs differ in shape between the two panels?

Agent's Pairwise Check



(c) exp i
Agent: different

(d) exp ii
Agent: same

Reasoning & Answer

Claude's reasoning excerpt:

"Let me create a montage of the original color crops enlarged for each pair, brightening the darker left crops..." presented with Figure (c) (d) tops

"...another careful look at the borderline cases by generating clean, smoothed silhouettes..." presented with Figure (c) (d) bottoms

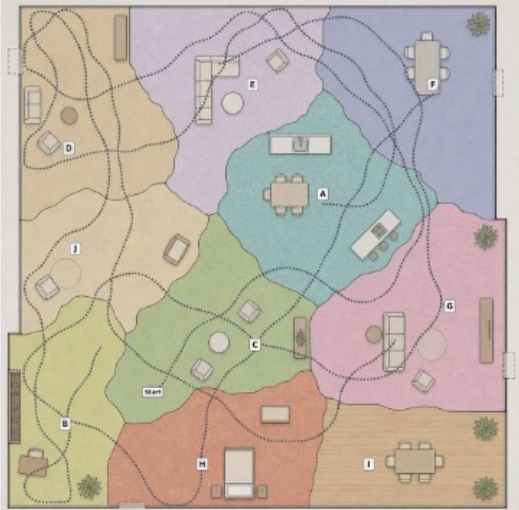
Analysis:

Realistic textures break the tools: the regenerated masks bake in shadows and panel creases from the dark, **badly scaled left crops, distorting the true contours**. The errors net out to one wrong "same".

Truth: 15
Agent: 14

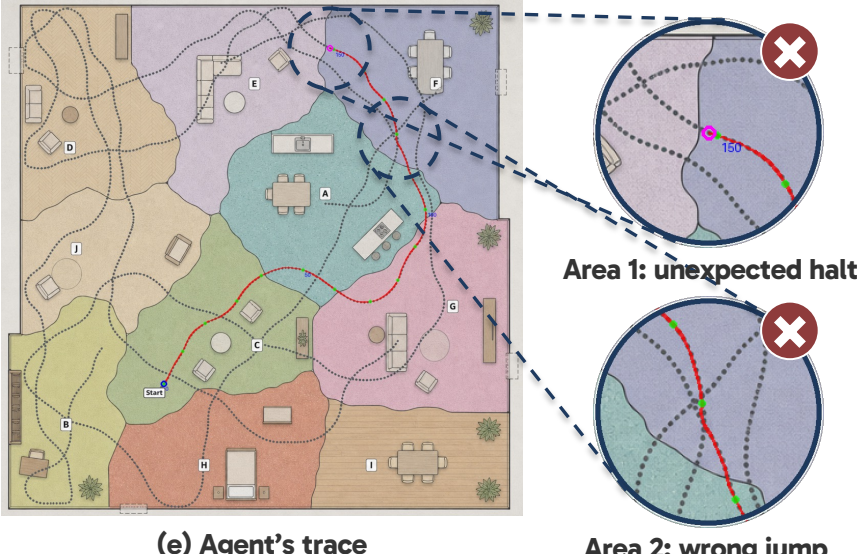
Failure 2. Weak visual perception prevents the agent from correcting its tool-call errors.

Color Zone Sequencing



List the region labels the curve visits in order.

Agent's Dot-center Trace



(e) Agent's trace

Area 1: unexpected halt
Area 2: wrong jump

Reasoning & Answer

GPT5.5's reasoning excerpt:

"The trace reaches a tight knot near the E/F boundary... the next dot is partly merged with a boundary/overlap... sends the curve across E... where the visible strand ends." presented with Figure (e)

Analysis:

The tracer jumps strands at a crossing (Area 2), then halts early (Area 1). The agent **audits neither**.

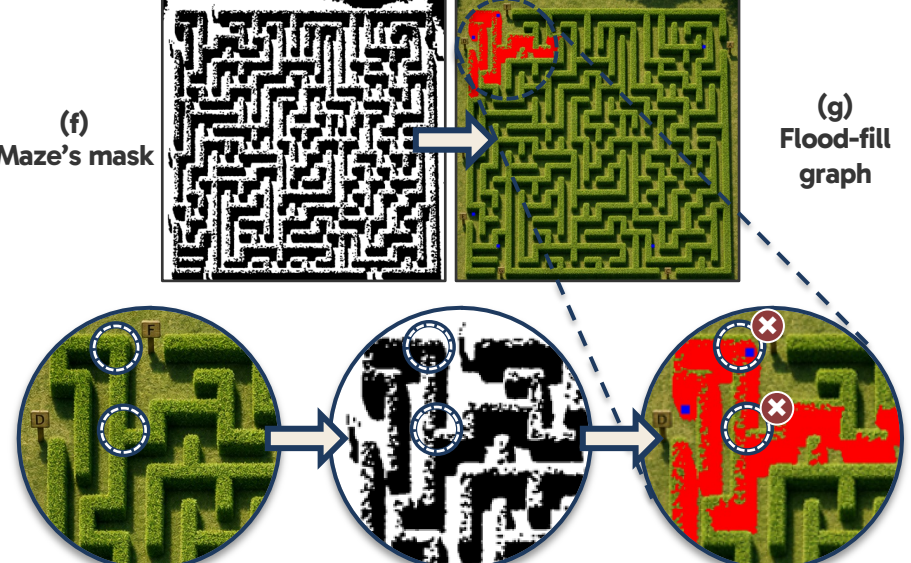
Truth: C, A, G, A, F, E, D, J, B, C, H, I, G
Agent: C, A, G, A, F, E

Maze Path Tracing



Which pair of openings is joined by a corridor?

Agent's Analysis Process



(f) Maze's mask (g) Flood-fill graph

Reasoning & Answer

Claude's reasoning excerpt:

"The cleaned mask reveals the maze structure well, with white representing walls and black showing the corridors." presented with Figure (f)

"...the red region traces actual maze corridors... never crossing hedge walls, and connects F's opening to D's opening..." presented with Figure (g)

Analysis:

The mask misreads a **shadowed hedge as corridor**, fusing D and F. The agent's perception **never catches the leak** and trusts the flawed output.

Truth: E-F
Agent: D-F